

High-resolution Coral Geodesy in the Solomon Islands

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The classical earthquake cycle is commonly described as alternating between long periods (decades to centuries) of interseismic locking and brief episodes (seconds) of coseismic rupture. However, increasingly dense geodetic observations from recent megathrust earthquakes reveal a more complex spectrum of transient deformation processes that challenge this binary framework. The New Georgia Group in the Solomon Islands provides a unique natural laboratory to investigate these processes, where the Woodlark Basin subducts beneath the Solomon Arc and has generated large megathrust earthquakes, including the 1936 Mw 7.9 and 2007 Mw 8.1 events.

The close proximity of the islands to the trench allows *Porites* corals to serve as high-resolution recorders of vertical ground motion. While coral morphology has long been used to identify coseismic uplift, we introduce a novel approach that combines coral morphology with stable isotope analysis ($\delta^{13}\text{C}$ and $\delta^{18}\text{O}$) to quantify relative sea-level (RSL) variations at annual resolution. We first assess the robustness of the relationship between coral water depth and $\delta^{13}\text{C}$ using 141 new samples collected across a range of depths formed within the same time interval. For depths between 170 and 110 cm below sea level, $\delta^{13}\text{C}$ exhibits a strong linear correlation with water depth ($R^2 = 0.982$), while shallower samples display a non-linear response.

We then apply this RSL proxy to a 692-sample coral time series spanning 1928–2012 and validate the reconstructed RSL against available tide-gauge records. The 2007 Mw 8.1 earthquake is clearly resolved, with coral morphology recording ~70 cm of coseismic uplift expressed as a pronounced die-down surface, accompanied by a $\delta^{13}\text{C}$ excursion exceeding 2‰. The 1936 Mw 7.9 event is similarly captured by a distinct $\delta^{18}\text{O}$ anomaly, with postseismic relaxation observed consistently along two independent drilling transects.

Beyond discrete coseismic signals, the record reveals multi-year to decadal periods of uplift and subsidence that we interpret as complex interseismic deformation. In particular, we identify intervals consistent with slow slip activity during 1955–1964, 1977–1986, and 1999–2002. These results demonstrate that stable isotope measurements in corals provide a powerful bridge between instrumental geodesy and paleoseismology, enabling a continuous, high-resolution view of subduction-zone deformation and stress evolution across the full earthquake cycle.